THE UNIVERSITY OF BRITISH COLUMBIA

The Impact of Bilingualism on the Developmental Trajectory of Vowel Harmony Sensitivity

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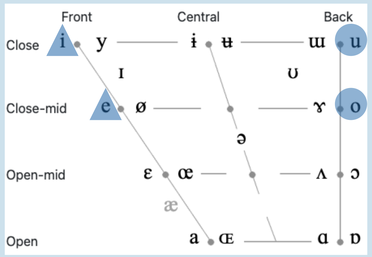
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Introduction

- **Vowel harmony (VH)** is a phonological process in which vowels within a word agree on certain phonetic features[1]
  - e.g., backness in Turkish
    çocuk-lar "children"
    ev-ler "houses"



- Disharmonies can signal word boundaries, making VH a potential cue for word segmentation[2]
- Infants learning VH languages (e.g., Turkish[3] and Akan[4]) distinguish harmonic vs. disharmonic words by 6 months, and use VH for word segmentation by 9-11 months[5]
- Recent evidence suggests an **early, broad sensitivity to VH**:
  - 4-month-old infants primarily exposed to non-VH languages prefer harmonic compared to disharmonic sequences[6]
  - 7-month-old English-learning infants (a non-VH language) can segment speech using VH cues[7]
- This sensitivity **attenuates without relevant input**:
  - By 7-8 months, infants exposed primarily to non-VH language(s) no longer prefer harmonic words[6,7], a pattern consistent with perceptual attunement
- Bilingual infants may follow a different developmental path[8]:
  - Experience with multiple languages could maintain broader perceptual sensitivity[9]
  - However, exposure to more than one non-VH language may also accelerate the down-weighting of VH as a cue

Does VH sensitivity decline in infants with no exposure to VH, and does exposure to more than one non-VH language change this trajectory?

Participants

- 35 mono- and 38 bilingual 6-month-old infants
- 32 mono- and 22 bilingual 10-month-old infants
- All infants had no to minimal (< 5%) exposure to VH

Methods – Central Fixation

Baseline Phase (testing preference)

- 4 fixed-length trials with a disyllabic word repeated 16 times, with 500-ms interstimulus interval (ISI)

ke-mi ***no-gu*** **vs.** ***ki-mo*** ***nu-ge***

Familiarization Phase

- 108-s continuous speech stream (45 repetitions) with no acoustic, prosodic, or statistical cues to word boundaries

de-bi-do-tu-be-pi-to-pu

Test Phase (testing segmentation)

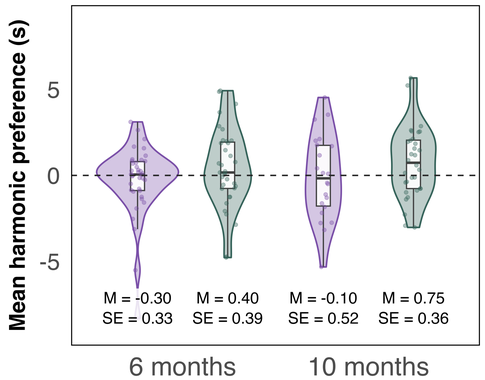
- 8 infant-controlled trials presenting a disyllabic sequence from the familiarization stream with 500-ms ISI

to-pu ***de-bi*** **vs.** ***bi-do*** ***tu-be***

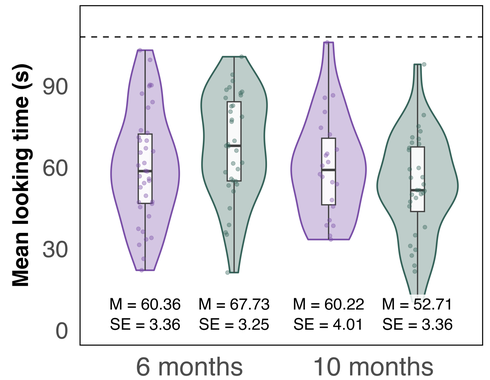
Preliminary Results

Bilinguals Monolinguals

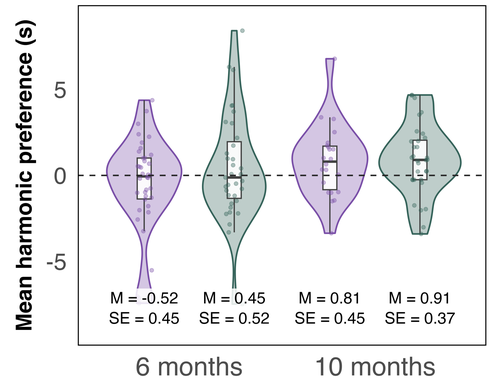
At baseline, harmony interacts with bilingualism (β = 0.18, p < .001): monolinguals preferred harmony (β = -0.22, p = .021), but no evidence for a significant preference in bilinguals (β = 0.14, p = .058).



In familiarization, 10-month-olds looked less than 6-month-olds (β = -3.63, p = .042), and age interacts with bilingualism (β = 3.83, p = .032): effect was primarily driven by monolinguals (β = -7.51, p = .002).



At test, 10-month-olds listened longer to harmonic sequences (β = 0.44, p = .021), but this effect was only significant if bilingual and monolingual infants were combined.



Summary & Discussion

- This is one of the first studies to apply the perceptual attunement framework to structural cues like vowel harmony.
- No clear evidence that infants use vowel harmony to segment words: no group reliably preferred harmonic words or disharmonic part-words at test.
- Consistent with prior work, English-learning monolinguals prefer harmonic word forms at baseline, indicating this bias can exist without vowel harmony input. We found no evidence of a decline in harmonic preference between 6 and 10 months.
- Bilinguals didn't show a significant preference at baseline. Exposure to multiple non-harmonic languages may provide stronger evidence that vowel harmony isn't a useful cue, accelerating attunement away from it.

References: [1] Gordon, M. K. & Fiddler, M. (2024). Vowel harmony: Statistical perspectives on typological distribution. [2] Vroomen, J., Tuomainen, J., & de Gelder, B. (1998). The roles of word stress and vowel harmony in speech segmentation. [3] Altan, A., Kaya, U., & Hohenberger, A. (2016). Sensitivity of Turkish infants to vowel harmony in stem-suffix sequences: Preference shift from familiarity to novelty. [4] Omane, P. O., Benders, T., & Boll-Avetisyan, N. (2024). Vowel harmony preferences in infants growing up in multilingual Ghana (Africa). [5] Omane, P.O., Boll-Avetisyan, N. and Benders, T. (2025). Tongue root harmony cues for speech segmentation in multilingually raised infants learning languages with and without vowel harmony in Ghana (Africa). [6] Solá-Llonch, E., & Sundara, M. (2025). Young infants' sensitivity to precursors of vowel harmony is independent of language experience. [7] Mintz, T. H., Walker, R. L., Welday, A., & Kidd, C. (2018). Infants' sensitivity to vowel harmony and its role in segmenting speech. [8] Werker, J. (2012). Perceptual foundations of bilingual acquisition in infancy. [9] Petitto, L. A., Berens, M. S., Kovelman, I., Dubins, M. H., Jasinska, K., & Shalinsky, M. (2012). The "Perceptual Wedge Hypothesis" as the basis for bilingual babies' phonetic processing advantage: New insights from fNIRS brain imaging.

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Sound Files

Baseline Phase (testing preference)

https://helenshiyang.github.io/assets/audio/feelharmonic/har_MH_kemi_16.mp3

ke-mi ***no-gu*** **vs.** ***ki-mo*** ***nu-ge***

Familiarization Phase

https://helenshiyang.github.io/assets/audio/feelharmonic/stream45_MH_audioramp.mp3

de-bi-do-tu-be-pi-to-pu

Test Phase (testing segmentation)

https://helenshiyang.github.io/assets/audio/feelharmonic/word_MH_topu_18.mp3

to-pu ***de-bi*** **vs.** ***bi-do*** ***tu-be***

Replication Study

Familiarization Phase

108-s continuous speech stream (45 repetitions) with no acoustic, prosodic, or statistical cues to word boundaries

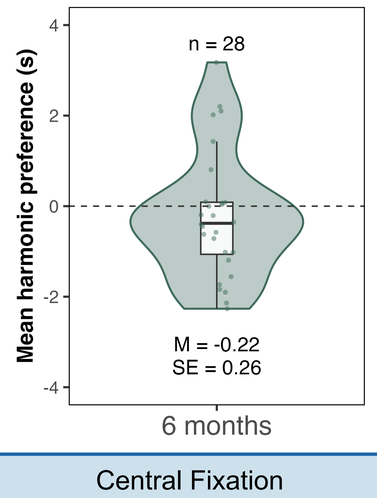
di-te-pu-bo-bi-de-tu-po

Test Phase

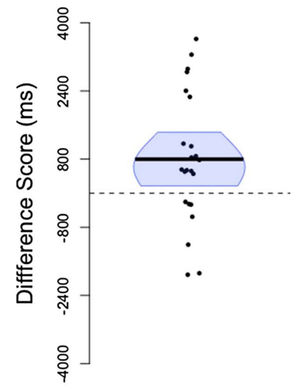
8 infant-controlled trials presenting a disyllabic sequence from the familiarization stream with 500-ms ISI

di-te ***tu-po*** **vs.** ***po-di*** ***de-tu***

In our replication, 6-month-olds listened longer to disharmonic part-words at test ($\beta = 0.15$, $p = .042$)



In the original study, 7-month-olds listened longer to harmonic words at test ($F(1,22) = 7.548$, $p = .012$; Mintz et al., 2018)



Central Fixation

Head-turn Preference